BA-TOS: from SIBGRAPI 2026 to a journal version

status and decisions

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Where we started: the SIBGRAPI 2026 result

Data	Crops	Specimens	Crop-macro F1@0.50	Without safeguard
EFRGP01, 2 scans, 200^3	12	1	**0.906** $\pm$ **0.210** (min 0.260)	0.824 $\pm$ 0.367 (min 0.012)

Three open issues

1. **Safeguard is retrospective:** the support-collapse gate was added after seeing crop05 fail.
2. **Parameters:** dozens of hand-set constants, never audited.
3. **No external validation:** one specimen, the same 12 crops used for design and evaluation.

Source: `results/batos_sibgrapi/crop_macro_summary.csv` (seed-ignore, no size filter, sample SD)

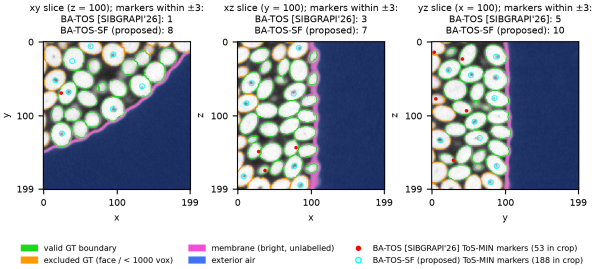
Step 1: reproduce before changing anything

- The camera-ready code was lost with the Linux machine; the E-series pipeline was rebuilt from recovered scripts.
- Run verbatim, it reproduces **Table II on all 12 crops within** ± 0.0005.
- Deterministic: 4 reruns bit-identical; SHA-256 of every script recorded; tag `batos-paper-v2-frozen`.

Why it matters

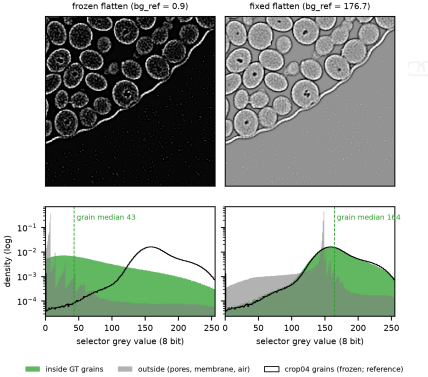
Every later number is a controlled change against this frozen reference, not against a memory of the paper.

Root cause of the crop05 failure: a flatten offset



Membrane (magenta), exterior air (blue, 51 % of the crop); markers:

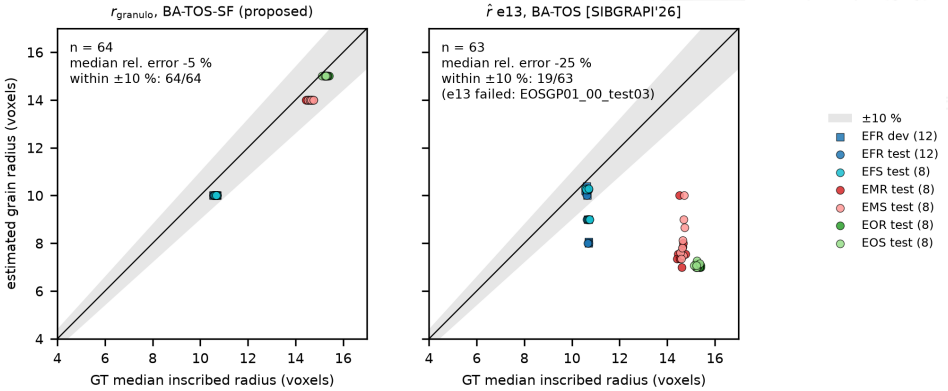
SIBGRAPI (red) vs BA-TOS-SF (cyan)



Background reference: 0.9 (frozen) vs 176.7 (fixed)

- Air at value 0 drags $\mathrm{median}(\mathit{blur} > 0)$ to 0.9; **one-line fix:** $\mathrm{median}(\mathit{blur} \geq 1)$.
- The safeguard then never fires; crop-macro F1@0.50 **0.824** $\rightarrow$ **0.976** (12 crops).

Scale: a granulometric estimator with no parameter



- r_{granulo} = argmax of the pattern spectrum of the specimen foreground: **64/64 crops within ±10 %** of the GT radius.
- The SIBGRAPI estimator (e13): 19/63 within ±10 %; it **halves the radius** on the larger EM / EO grains.

BA-TOS-SF (v3): what changed

- Scale from r_{granulo}; every size constant follows $s = r_{\mathrm{granulo}}/10$.
- Thresholds from specimen-only statistics (Otsu inside the specimen).
- Absolute grey-level guards removed.
- No safeguard needed: the gate never fires.

93 constants	**SIBGRAPI**	**BA-TOS-SF**
Fixed in voxels (size, volume, count)	43	**0**
Ad-hoc thresholds	6	**0**
Absolute grey-level guards	3	**0**
Dimensionless (kept)	41	41
Gate fires (52 test crops)	30	**0**

Size constants scale as s, s^3 or $1/s^3$; on the dev crops $s = 1$.

Source: `docs/V3_SCALE_FREE.md (full constant table); paper/tables/test_v3_tables.tex`

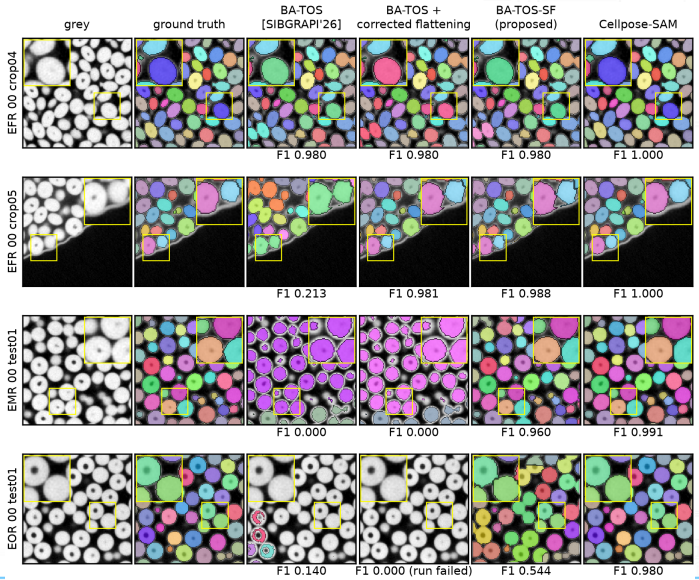
Held-out validation: 52 crops, 6 specimens

Specimen	n	$\hat{r}$ / r_{in}	SIBGRAPI'26	+ fixed flattening	BA-TOS-SF
EFR	12	10 / 10.6	0.879 ± 0.223	0.956 ± 0.048	$\mathbf{0.985 \pm 0.011}$
EFS	8	10 / 10.6	0.882 ± 0.155	0.968 ± 0.043	$\mathbf{0.974 \pm 0.044}$
EMR	8	14 / 14.5	0.303 ± 0.439	0.250 ± 0.416	$\mathbf{0.980 \pm 0.027}$
EMS	8	14 / 14.7	0.432 ± 0.465	0.479 ± 0.474	$\mathbf{0.986 \pm 0.020}$
EOR	8	15 / 15.3	0.293 ± 0.362	0.238 ± 0.434	$\mathbf{0.754 \pm 0.240}$
EOS	8	15 / 15.3	0.265 ± 0.429	0.365 ± 0.499	$\mathbf{0.754 \pm 0.214}$
External (5 spec.)	40	–	0.435 ± 0.435	0.460 ± 0.473	$\mathbf{0.890 \pm 0.178}$
All	52	–	0.538 ± 0.437	0.575 ± 0.465	$\mathbf{0.912 \pm 0.161}$

- Protocol pre-registered: crop list committed before cutting (GT-free rule, fixed seed); nothing tuned on test.
- BA-TOS-SF fixes the EM specimens; the large-grained EO specimens remain the weak point (0.75).

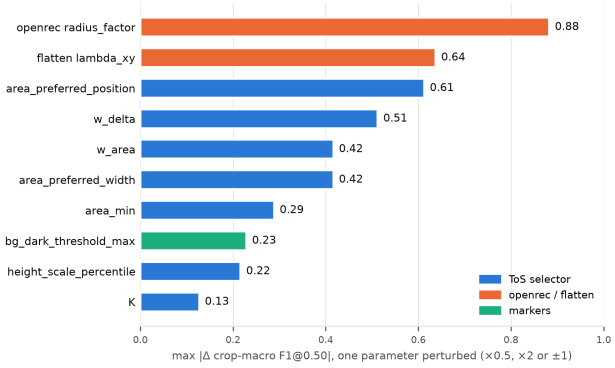
Crop-macro F1@0.50, seed-ignore, no size filter. Source: `results/test_v3/`

Qualitative: 4 crops × 6 columns



Parameter audit: 96 constants, 25 influential

- 96 method constants swept one at a time (12 dev crops); **25** move F1 by > 0.01.
- Selector weights sit on a **narrow ridge**: halving or doubling one drops F1 by 0.4–0.6.
- `w_support` is **dead code**: $\Delta = 0.000$.



Source: `results/sensitivity/ranking.csv` (top 10 method parameters, BA-TOS_sel route)

The honest comparison: Cellpose-SAM, untuned

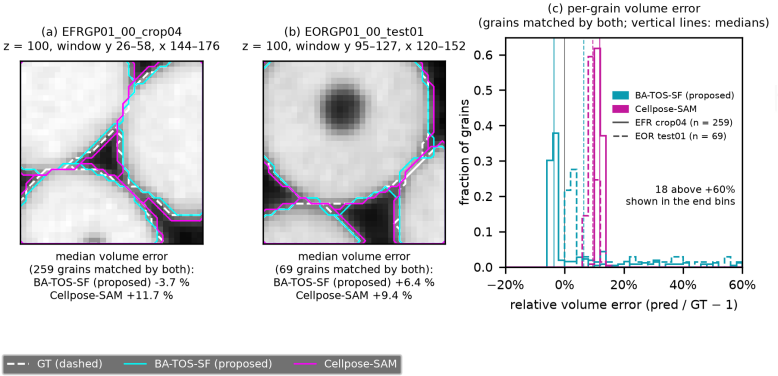
Specimen	`cpsam_v2`	BA-TOS-SF
EFR	**0.999**	0.977
EFS	**0.999**	0.964
EMR	**0.998**	0.966
EMS	**0.999**	0.970
EOR	**0.996**	0.735
EOS	**0.998**	0.727
All 52	**0.998**	0.896

Crop-macro F1@0.50, no-marker evaluation (same rule for both)

- Default settings, one config, no fine-tuning; all four Cellpose 4 models agree within 0.002.
- **Robustness:** at 5 % added noise BA-TOS-SF collapses (0.13 / 0.06 / 0.00 on 3 crops), `cpsam_v2` $\geq$ 0.977; at 2$\times$ scale 0.963 vs 0.900.
- **No training-data leakage:** the documented Cellpose-SAM set is 19 cell/nuclei datasets, 2D only (plus 98 natural photos, e.g. rocks); no micro-CT, granular data or Zenodo 8014905.
- June 2026 models have no published list, but the documented `cpsam` scores the same (0.999).

Source: `results/cellpose/REPORT.md §2, §7, §7b; tables.md`

Where BA-TOS still differs: boundaries



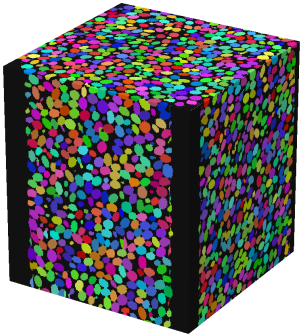
- Cellpose-SAM: consistent **+6 to +14 %** volume bias (about one voxel outside the GT).
- BA-TOS-SF: median −3.7 % on crop04, but a **heavy tail of merged grains** (46/259 grains above +20 %).
- Hybrid (Cellpose markers + morphological boundary) is running.

Decisions for today

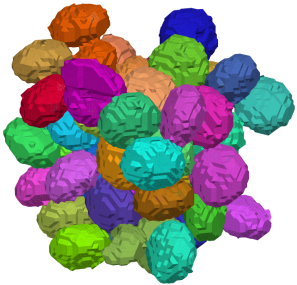
1. **Thesis:**
   (A) diagnostic + scale-free method + honest benchmark, or
   (B) hybrid for grain analysis (Cellpose markers, morphological boundaries).
2. **Venue** for the journal version.
3. **When to involve** Yukiko / Nicolas, and Pinzón / Viggiani: how was the GT made?
4. **Still running:** full volume, v3c, hybrid, tracking.

Backup: full-volume tiled BA-TOS-SF

(a) stitched BA-TOS-SF (proposed) block (600^3 core, 3×3×3 tiles), faces cut 30 voxels inside (540^3 shown)



(b) 55 grains with centroid in a central 100^3 sub-block (marching cubes)



- Central $3 \times 3 \times 3$ tiles of EFRGP01_00: 600^3 core, halo 20, orphan-rescue stitching.
- 9,753 instances; F1@0.50 = 0.865 on the 6,005 valid GT grains (seed-ignore, no filter).

The end

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QUESTIONS AND COMMENTS?



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